厦门兰阶教育 2022 年学科奥赛•物理珍题集训

模拟试卷一（卷Ⅰ）　命题者：苏国珍教授

学　校_____

姓　名_____

条码粘贴处

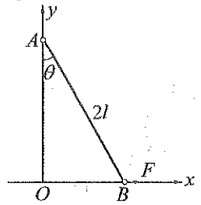
请在各题目的答题区域内作答，超出黑色矩形边框限定区域的答案无效

一.（34 分）如图，质量为 m，长为 $2l$ 的匀质细杆，其两端 A、B 分别被约束在沿水平和竖直方向的光滑导轨上运动。设重力加速度为 g。

（1）若杆末端 B 受到沿水平方向的外力 F 的作用，使杆在 $\angle OAB=\theta$ 的位置保持静止，求作用力 F 的大小；[8 分]

（2）通过水平力 F 的作用，使杆的 B 端以恒定速度 v_0 向左运动，求当 $\angle OAB=\theta$ 时，作用力 F 的大小；[14 分]

（3）续第（2）问，当 $\angle OAB=\theta$ 时突然撤去 F，求撤去 F 后瞬间，杆末端 B 的加速度。[12 分]



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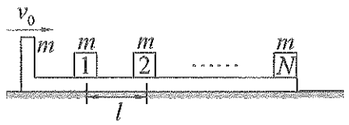
二.（40 分）如图，左端带有挡板的长木板静置在光滑水平地面上，N 个相同的小滑块（可视为质点）等间距地静置在木板上，从左到右依次编号为1、2 … N，其中第 N 个的滑块恰好处于木板的右端点。已知每个滑块和木板的质量均为 m，相邻两滑块相距为 l，滑块与木板间的动摩擦系数为 μ，重力加速度为 g。今通过外力作用使木板瞬间获得一个水平向右的初速度 v_0。

（1）假设挡板与滑块之间、以及滑块与滑块之间的碰撞均为完全弹性碰撞，且碰撞时间极短，并设 $N=10$, $v_0=6\ \mathrm{m/s}$，$l=0.2\ \mathrm{m}$，$\mu=0.5$，$g=10\ \mathrm{m/s^2}$。求最终各物滑块的速度大小（相对于地面）和位置（相对于木板）；[16 分]

（2）假设挡板与滑块之间、以及滑块与滑块之间的碰撞均为完全非弹性碰撞，且碰撞时间极短。并设 $N=10$, $v_0=10\ \mathrm{m/s}$，$l=0.1\ \mathrm{m}$，$\mu=0.25$，$g=10\ \mathrm{m/s^2}$。求最终各物滑块的速度大小（相对于地面）和位置（相对于木板）。[24 分]



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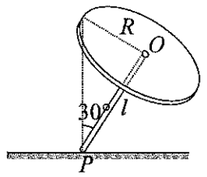
三.（44 分）一质量为m，半径为R的匀质圆盘与质量为$m/2$，长度为$l=\sqrt{3}R$的杆的一端固连于圆心O处，杆与盘面垂直，杆的另一端P置于粗糙的水平地面内。初始杆与竖直线的夹角为$30°$，系统从静止释放。

（1）假设杆与水平面间保持无滑动，求杆与竖直线夹角为$\theta\,(30°\le\theta<60°)$时，杆转动的角速度和角加速度。[10 分]

（2）在圆盘触地之前，杆与水平面间能否始终保持无滑动？若能，求出所需的最小摩擦系数；若不能，请说明理由。[14 分]

（3）若初始圆盘不是静止，而是绕轴PO以角速度ω高速旋转，则释放后系统还将绕过P点的竖直轴转动（进动）。假设杆与水平面间保持无滑动，求系统进动的角速度。[10 分]

（4）在第（3）问的条件下，求保持杆与水平面间无滑动所需的最小摩擦系数。[10 分]



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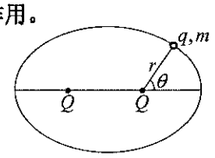
四.（40 分）如图所示，一细钢丝弯成极坐标方程为

$$r=\frac{p}{1+\varepsilon\cos\theta}$$

的椭圆，其中p、ε为正的常量，$0<\varepsilon<1$。在椭圆的两个焦点上各固定电量为Q的点电荷，另一质量为m，带电量为q的小球串在钢丝上，与钢丝绝缘，可沿钢丝无摩擦滑动。不考虑重力和其它阻力的作用。

（1）求小球可能的平衡位置，并判断各平衡位置的稳定性；[10 分]

（2）求小球在各稳定平衡位置附近作微小振动的周期。[30 分]



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五.（42 分）如图所示，两半径分别为 r_1 和 r_2 $(r_2 > r_1)$ 的理想导体薄球壳同心放置。内球壳接地，并通过电阻可忽略的导线、阻值为 R（R 较大）的电阻器和开关 K 与外球壳相连。在距离球心 O 为 a $(a > r_2)$ 处放置一个电量为 Q 的点电荷。开始时，开关 K 断开，外球壳所带总电量为零，系统已达到静电平衡。$t = 0$ 时刻，闭合开关 K。求

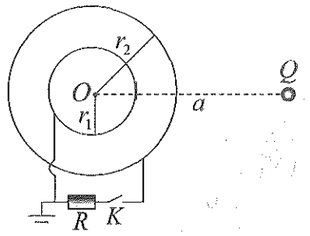
（1）开关 K 闭合之前内球壳所带的电量、外球壳的电势以及系统的静电能（不含点电荷 Q 的自能）；[12 分]

（2）开关 K 刚闭合后瞬间，通过电阻器 R 的电流强度；[4 分]

（3）开关 K 闭合足够长时间后，两球壳所带的电量；[4 分]

（4）开关 K 闭合后任意 t 时刻通过电阻器 R 的电流强度；[16 分]

（5）开关 K 闭合后足够长时间内，电阻器 R 消耗的总焦耳热。[6 分]



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六.（40 分）克劳修斯不等式是热力学中的一个重要结论。设工作物质在一个循环中与温度分别为T_1、T_2、$\cdots T_n$的n个恒温热源产生热交换，从各热源吸收的热量分别为Q_1、Q_2、$\cdots Q_n$（如果从某个热源吸收的热量为负值，则表示工作物质向该热源放出热量），则

$$\sum_{i=1}^{n}\frac{Q_i}{T_i}\le 0$$

如果循环是可逆的，则以上"≤"取等号"="。上述表达式称为克劳修斯不等式。本题将讨论克劳修斯不等式的一些应用。

（1）传统的教科书上都是以热力学第二定律作为基本出发点，由此证明卡诺定理和克劳修斯不等式，这个证明过程是较为复杂的。然而如果把证明过程的方向倒过来，则可能要简单得多。试以克劳修斯不等式作为已知的基本结论，证明卡诺定理以及热力学第二定律的开尔文描述和克劳修斯描述。[14 分]

（2）假设某热机在循环中与多个恒温热源交换热量，在工作物质从其中吸收热量的热源中，热源的最高温度为T_1，在工作物质向其中放出热量的热源中，热源的最低温度为T_2。热机的效率定义为$\eta = W_{净}/Q_{吸}$，其中$W_{净}$表示循环过程中工作物质对外所做的净功，$Q_{吸}$表示循环中工作物质在各吸热过程中所吸收的总热量。试根据克劳修斯不等式证明，$\eta \le 1 - T_2/T_1$。[12 分]

（3）有N个的物体，初始温度分别为T_{01}、T_{02}、$\cdots$、T_{0N}，热容量均为C，且为常量。以这N个物体为热源（除此之外无其它热源），通过工作物质的循环过程使系统对外做功。试利用克劳修斯不等式求当N物体达到相同温度时系统对外所做的最大功$W_{\max}$。[14 分]

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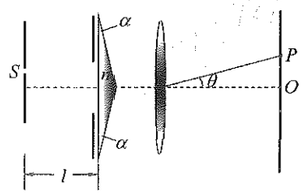
七.（40 分）如图所示，双棱镜两底角为α ($\alpha \ll 1$)，折算率为n。在双棱镜左边距离棱镜l处放置一狭缝S，S靠近双棱镜的光轴，但未必在光轴上。

（1）从狭缝S发出的某一光线以入射角i（i不一定很小）从左侧入射，求光线经过棱镜折射之后的偏转角。[10 分]

（2）在棱镜的左边放置一光阑，使得进入棱镜的所有光线的入射角都很小，求狭缝S经双棱镜折射而形成的两个虚像的位置；[10 分]

（3）在双棱镜后面放置一个焦距为f的理想凸透镜，其光轴与双棱镜的光轴重合，在透镜焦平面内有一屏幕，与光轴相交于O点，用θ表示屏幕上某一点P到透镜中心的连线与光轴的夹角。今用波长为λ的单色光照射狭缝，则屏幕上将看到干涉条纹。假设狭缝刚好位双棱镜的轴线上，试求P点为明纹中心时θ满足的条件；[8 分]

（4）假设狭缝不在光轴上，而是与光轴相距为b（但仍靠近光轴），其它条件与第（3）问的相同，再求P点为明纹中心时θ满足的条件。[12 分]



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八.（40 分）在实验室参照系中，一静止质量为 M_0 的飞弹以速度 v 高速飞行，在 $t=0$ 时刻被引爆，爆炸成 N 个弹片。已知在质心参照系中，这 N 个弹片具有相同的质量，速度大小刚好均为 v，且各向同性地分布在同一个平面内。考虑相对论效应。

（1）求在实验室参照系中，单个弹片可能的最大动能；[10 分]

（2）当 N 足够大时，各相邻弹片相连构成的曲线可视为光滑曲线。求在实验室参照系中，t 时刻该曲线的方程（统一以飞弹被引爆点为坐标原点，飞行方向为 x 轴正方向建立直角坐标系）；[12 分]

（3）续第（2）问，求在实验室参照系中测，上述曲线上弹片的最大数密度（曲线上单位长度的弹片数）。[18 分]

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中学生物理奥赛培训

模拟试卷（一）解答

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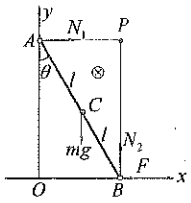
福建 厦门 2022.6

题一.

(1) $\begin{cases} N_1 - F = 0 \\ N_2 - mg = 0 \\ \tau_C = N_1 l\cos\theta + Fl\cos\theta - N_2 l\sin\theta = 0 \end{cases}$

$F = \frac{1}{2}mg\tan\theta$



另解：

$\tau_P = mgl\sin\theta - F\cdot 2l\cos\theta = 0$

$F = \frac{1}{2}mg\tan\theta$

(2)

杆作平面平行运动。$\vec{a}_C, \beta \Rightarrow F$　　$\beta = ?\ \vec{a}_C = ?$

$x = 2l\sin\theta$

$v_0 = -\frac{dx}{dt} = -\frac{dx}{d\theta}\frac{d\theta}{dt} = 2l\omega\cos\theta\ (\omega = -\frac{d\theta}{dt})$

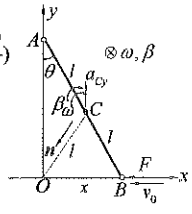
$\omega = \frac{v_0}{2l\cos\theta}$

$\beta = \frac{d\omega}{dt} = \frac{d\omega}{d\theta}\frac{d\theta}{dt} = -\omega\frac{d\omega}{d\theta} = -\frac{v_0^2\sin\theta}{4l^2\cos^3\theta}$

$a_{Cx} = \frac{1}{2}a_B = 0$

$a_{Cn} = -a_{Cy}\cos\theta = l\omega^2 \Rightarrow$

$a_{Cy} = -\frac{\omega^2}{\cos\theta} = -\frac{v_0^2}{4l\cos^3\theta}$

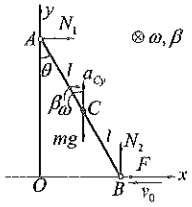


C 绕 O 点作圆周运动

$F = ?$

$\begin{cases} N_1 - F = ma_{Cx} \\ N_2 - mg = ma_{Cy} \\ Fl\cos\theta + N_1 l\cos\theta - N_2 l\sin\theta = \frac{1}{12}m(2l)^2\beta \\ a_{Cx} = 0,\ a_{Cy} = -\frac{v_0^2}{4l\cos^3\theta},\ \beta = -\frac{v_0^2\sin\theta}{4l^2\cos^3\theta} \end{cases}$

$F = \frac{1}{2}mg(1 - \frac{v_0^2}{3gl\cos^3\theta})\tan\theta$



$\beta = -\frac{v_0^2\sin\theta}{4l^2\cos^3\theta}$

$a_{Cx} = 0,\ a_{Cy} = -\frac{v_0^2}{4l\cos^3\theta}$

另解：

直角 ΔPAB 视为一个刚体。

杆相对于 P 点作定轴转动。

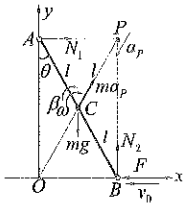
P 点的加速度不等于零，但 P 做直线运动，加速度方向始终沿 PO 方向。

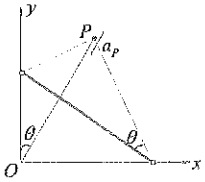
惯性力产生的对 P 点的力矩为零。

$\begin{cases} F(2l)\cos\theta - mgl\sin\theta = J_P\beta \\ \beta = -\frac{v_0^2\sin\theta}{4l^2\cos^3\theta},\ J_P = \frac{1}{12}m(2l)^2 + ml^2 \end{cases}$

$F = \frac{1}{2}mg(1 - \frac{v_0^2}{3gl\cos^3\theta})\tan\theta$

$\beta = -\frac{v_0^2\sin\theta}{4l^2\cos^3\theta}$





另解：能量法：

$E_p = mgl\cos\theta$

P为瞬心

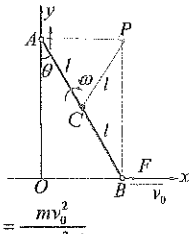
$\omega = \frac{v_0}{PB} = \frac{v_0}{2l\cos\theta}$

$E_k = \frac{1}{2}J_P\omega^2 = \frac{1}{2}[\frac{1}{12}m(2l)^2 + ml^2](\frac{v_0}{2l\cos\theta})^2 = \frac{mv_0^2}{6\cos^2\theta}$

$E = E_p + E_k = mgl\cos\theta + \frac{mv_0^2}{6\cos^2\theta}$

$Fv_0 = \frac{dE}{dt} = \frac{dE}{d\theta}\frac{d\theta}{dt} = -\omega\frac{dE}{d\theta} = \frac{1}{2}mg\tan\theta(1 - \frac{v_0^2}{3gl\cos^3\theta})v_0$

$F = \frac{1}{2}mg(1 - \frac{v_0^2}{3gl\cos^3\theta})\tan\theta$



(3)

$$\begin{cases} -mgl\sin\theta = J_P\beta \\ J_P = \frac{1}{12}m(2l)^2 + ml^2 \end{cases} \Rightarrow \beta = -\frac{3g\sin\theta}{4l}$$

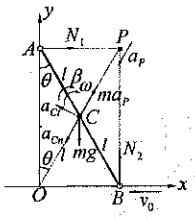
$$a_{Cn} = l\omega^2 = \frac{v_0^2}{4l\cos^2\theta},\ a_{Ct} = l\beta = -\frac{3g\sin\theta}{4}$$

$$a_{Cx} = -a_{Ct}\cos\theta - a_{Cn}\sin\theta$$

$$= \frac{1}{4}(3g\cos\theta - \frac{v_0^2}{l\cos^2\theta})\sin\theta$$

$$a_{Bx} = 2a_{Cx} = \frac{1}{2}(3g\cos\theta - \frac{v_0^2}{l\cos^2\theta})\sin\theta$$

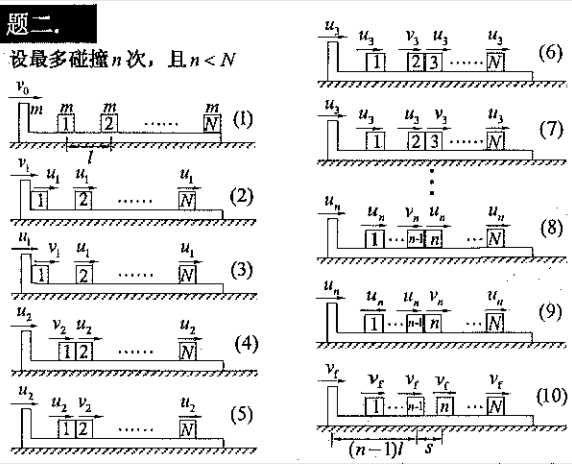
$$\omega = \frac{v_0}{2l\cos\theta}$$



C 绕 O 点作圆周运动

题二.

(1) 设最多碰撞 n 次，且 $n < N$



(1)→(2) 摩擦力所做总功：

$-N\mu mgl$

(2)→(9) 摩擦力所做总功：

$-(n-1)\mu mgl$

(9)→(10) 摩擦力所做总功：

$-\mu mgs$

(1)→(10) 功能原理：

$-N\mu mgl - (n-1)\mu mgl - \mu mgs$

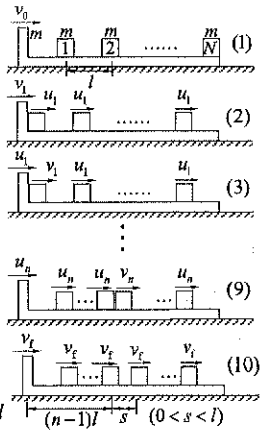
$= (N+1)\times mv_f^2/2 - mv_0^2/2$ (1)

(1)→(10) 动量守恒：

$mv_0 = (N+1)mu_f$ (2)

(1)(2) $\Rightarrow v_f = \frac{v_0}{N+1}$

$$s = \frac{N}{N+1}\frac{v_0^2}{2\mu g} - (N+n-1)l \quad (0 < s < l)$$



$0 < s < l \Rightarrow$

$$\frac{N}{N+1}\frac{v_0^2}{2\mu gl} - N < n < \frac{N}{N+1}\frac{v_0^2}{2\mu gl} - N + 1$$

$6.36 < n < 7.36$

$n = 7$

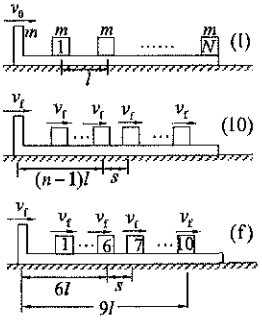
$v_f = \frac{6}{11} \approx 0.545$ m/s

$s = \frac{4}{55} \approx 0.073$ m

滑块1～6、8～10到挡板的距离分别为0.2m、0.4m…1.8m；

滑块7到挡板的距离为1.273m。

$$v_f = \frac{v_0}{N+1},\ s = \frac{N}{N+1}\frac{v_0^2}{2\mu g} - (N+n-1)l$$



(2)

设最多碰撞 n 次，且 $n < N$

$mv_0 = (N+1)mv_f$

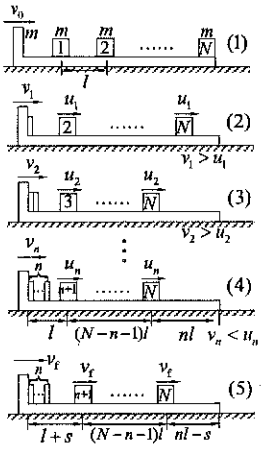
$v_f = \frac{v_0}{N+1} = \frac{10}{11} \approx 0.909$ m/s

$(n+1)mv_n + (N-n)mu_n = mv_0$ (1)

$-(N-n)\mu mgs = \frac{1}{2}(N+1)mv_f^2$

$-\frac{1}{2}(n+1)mv_n^2 - \frac{1}{2}(N-n)mu_n^2$ (2)

未知数：v_n, u_n, s, n



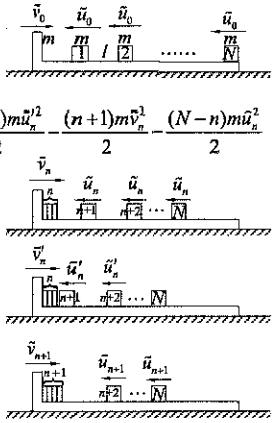
在质心参照系中求解。

$$v_C = \frac{v_0}{N+1},\ \tilde{u}_0 = \frac{v_0}{N+1},\ \tilde{v}_0 = \frac{Nv_0}{N+1}$$

$$\begin{cases} -\mu(N-n)mgl = \frac{(n+1)m\tilde{v}_n'^2}{2} + \frac{(N-n)m\tilde{u}_n'^2}{2} - \frac{(n+1)m\tilde{v}_n^2}{2} - \frac{(N-n)m\tilde{u}_n^2}{2} \\ (n+1)m\tilde{v}_n - (N-n)m\tilde{u}_n = 0 \\ (n+1)m\tilde{v}_n' - (N-n)m\tilde{u}_n' = 0 \\ \tilde{u}_{n+1} = \tilde{u}_n' \end{cases}$$

$$\tilde{u}_{n+1}^2 = \tilde{u}_n^2 - (n+1)\frac{2\mu gl}{N+1}$$

$$\tilde{v}_n = -\frac{N-n}{n+1}\tilde{u}_n$$



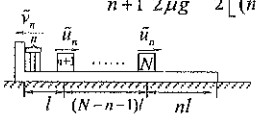
$$\tilde{u}_0=\frac{v_0}{N+1}\Rightarrow\cdots\ \tilde{u}_n^2=\frac{v_0^2}{(N+1)^2}-\frac{n(n+1)}{N+1}\mu gl$$

设最多碰撞 n 次，且 $n<N$

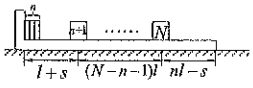
且最终滑块不会离开木板：$s>0$ 且 $nl-s>0$

$$-(N-n)\mu mgs=-\frac{(n+1)m\tilde{v}_n^2}{2}-\frac{(N-n)m\tilde{u}_n^2}{2}=-\frac{(N-n)(N+1)}{2(n+1)}m\tilde{u}_n^2$$

$$s=\frac{N+1}{n+1}\frac{\tilde{u}_n^2}{2\mu g}=\frac{1}{2}\left[\frac{1}{(n+1)(N+1)}\frac{v_0^2}{\mu g}-nl\right]$$



第 n 次碰撞后



终态，系统在质心系中静止

($\tilde{v}_n$ 向左，$\tilde{u}_n$ 向右)

$$\tilde{u}_0=\frac{v_0}{N+1},\ \tilde{u}_{n+1}^2=\tilde{u}_n^2-(n+1)\frac{2\mu gl}{N+1},\ \tilde{v}_n=-\frac{N-n}{n+1}\tilde{u}_n$$

$$s>0\Rightarrow\frac{1}{(n+1)(N+1)}\frac{v_0^2}{\mu g}-nl>0$$

$$n^2+n-\frac{1}{N+1}\frac{v_0^2}{\mu gl}<0$$

$$n^2+n-36.7<0$$

$$-6.55<n<5.55$$

$$n=5$$

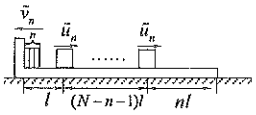
$$s=\frac{1}{2}\left[\frac{1}{6(N+1)}\frac{v_0^2}{\mu g}-5l\right]\approx 0.053\ \text{m}$$

$$5l-s=0.447>0$$

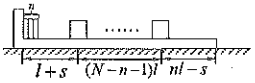
滑块1～5到挡板的距离为零；

滑块6～10到挡板的距离分别为：

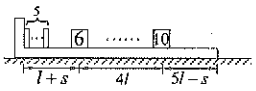
0.153 m, 0.253 m, ⋯0.553 m



第 n 次碰撞后

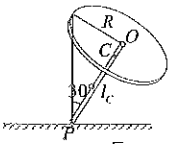


终态，系统在质心系中静止



$$s=\frac{1}{2}\left[\frac{1}{(n+1)(N+1)}\frac{v_0^2}{\mu g}-nl\right]$$

题三.

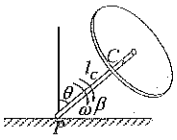


$OP=\sqrt{3}R$

(1) $l_C=\dfrac{m\sqrt{3}R+(m/2)(\sqrt{3}R/2)}{m+m/2}=\dfrac{5\sqrt{3}}{6}R$

$$J_P=\frac{1}{4}mR^2+m(\sqrt{3}R)^2+\frac{1}{3}\frac{m}{2}(\sqrt{3}R)^2=\frac{15}{4}mR^2$$

$$\begin{cases}\dfrac{3}{2}mgl_C\cos 30°=\dfrac{3}{2}mgl_C\cos\theta+\dfrac{1}{2}J_P\omega^2\\ l_C=\dfrac{5\sqrt{3}}{6}R,\ J_P=\dfrac{15}{4}mR^2\end{cases}$$



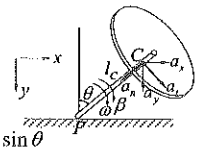
$$\omega=\sqrt{\frac{2\sqrt{3}g}{3R}\left(\frac{\sqrt{3}}{2}-\cos\theta\right)}$$

$$\beta=\frac{\mathrm{d}\omega}{\mathrm{d}t}=\omega\frac{\mathrm{d}\omega}{\mathrm{d}\theta}=\frac{1}{2}\frac{\mathrm{d}\omega^2}{\mathrm{d}\theta}=\frac{\sqrt{3}}{3}\frac{g}{R}\sin\theta$$

(2)

$$a_t=l_C\beta=\frac{5}{6}g\sin\theta$$

$$a_n=l_C\omega^2=\frac{5}{3}g\left(\frac{\sqrt{3}}{2}-\cos\theta\right)$$



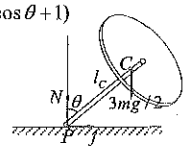
$$a_x=a_t\cos\theta-a_n\sin\theta=\frac{5}{6}g(3\cos\theta-\sqrt{3})\sin\theta$$

$$a_y=a_t\sin\theta+a_n\cos\theta=\frac{5}{6}g(-3\cos^2\theta+\sqrt{3}\cos\theta+1)$$

$$N=\frac{3}{2}mg-\frac{3}{2}ma_y=\frac{1}{4}mg(15\cos^2\theta-5\sqrt{3}\cos\theta+1)$$

$$f=\frac{3}{2}ma_x=\frac{5}{4}mg(3\cos\theta-\sqrt{3})\sin\theta$$

$$\tan\alpha=\frac{f}{N}=\frac{5(3\cos\theta-\sqrt{3})\sin\theta}{15\cos^2\theta-5\sqrt{3}\cos\theta+1}$$



$$\omega=\sqrt{\frac{2\sqrt{3}g}{3R}\left(\frac{\sqrt{3}}{2}-\cos\theta\right)},\ \beta=\frac{\sqrt{3}}{3}\frac{g}{R}\sin\theta,\ l_C=\frac{5\sqrt{3}}{6}R$$

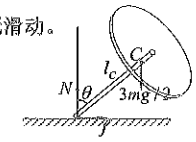
$$15\cos^2\theta-5\sqrt{3}\cos\theta+1=0$$

$$\cos\theta=\frac{\sqrt{3}}{30}(5+\sqrt{5}),\ \frac{\sqrt{3}}{30}(5-\sqrt{5})\Rightarrow\theta\approx 65.3°,\ 80.8°$$

$30°\le\theta\le 60°$范围内，$15\cos^2\theta-5\sqrt{3}\cos\theta+1>0$，$\tan\alpha$ 有限

只要μ足够大，杆能与水平面间保持无滑动。

$$\frac{\mathrm{d}\tan\alpha}{\mathrm{d}\theta}=\frac{5(51\cos^2\theta-31\sqrt{3}\cos\theta+12)}{(15\cos^2\theta-5\sqrt{3}\cos\theta+1)^2}$$



$$\left.\frac{\mathrm{d}\tan\alpha}{\mathrm{d}\theta}\right|_{\theta=\theta_m}=0\Rightarrow 51\cos^2\theta_m-31\sqrt{3}\cos\theta_m+12=0$$

$$\cos\theta_{m1}=\frac{31\sqrt{3}+2\sqrt{435}}{102}\approx 0.7309,\ \cos\theta_{m2}=\frac{31\sqrt{3}-2\sqrt{435}}{102}\approx 0.3219$$

$\theta_{m1}\approx 43.04°$，$\theta_{m2}\approx 71.22°$ (舍去)

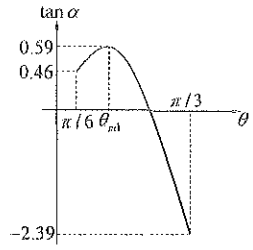
$$\tan\alpha=\frac{5(3\cos\theta-\sqrt{3})\sin\theta}{15\cos^2\theta-5\sqrt{3}\cos\theta+1}$$

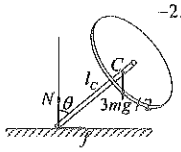
$$\tan\alpha|_{\theta=\pi/6}\approx 0.46$$

$$\tan\alpha|_{\theta=\theta_{m1}}\approx 0.59$$

$$\tan\alpha|_{\theta=\pi/3}\approx -2.39$$

$$\mu_{\min}=|\tan\alpha|_{\max}\approx 2.39$$





$$\tan\alpha=\frac{5(3\cos\theta-\sqrt{3})\sin\theta}{15\cos^2\theta-5\sqrt{3}\cos\theta+1},\ \theta_{m1}\approx 43.04°$$

(3)

$$M_P=\frac{3}{2}mgl_C\sin30^\circ=\frac{5\sqrt{3}}{8}mgR$$

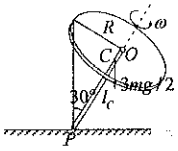
$$L\approx\frac{1}{2}mR^2\omega$$

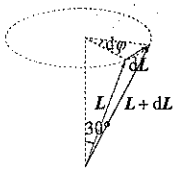
$$|\mathrm{d}\vec{L}|=L\sin30^\circ\cdot\mathrm{d}\varphi$$

$$\frac{|\mathrm{d}\vec{L}|}{\mathrm{d}t}=L\sin30^\circ\cdot\frac{\mathrm{d}\varphi}{\mathrm{d}t}=\frac{1}{2}L\Omega=\frac{1}{4}mR^2\omega\Omega$$

$$M_P=\frac{|\mathrm{d}\vec{L}|}{\mathrm{d}t}\Rightarrow\frac{5\sqrt{3}}{8}mgR=\frac{1}{4}mR^2\omega\Omega$$

$$\Omega=\frac{5\sqrt{3}}{2}\frac{g}{R\omega}$$





$$l_C=\frac{5\sqrt{3}}{6}R$$

(4)

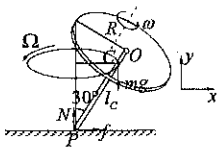
$$a_{Cx}=-(l_C\sin30^\circ)\Omega^2=-\frac{125\sqrt{3}}{16}\frac{g^2}{R\omega^2}$$

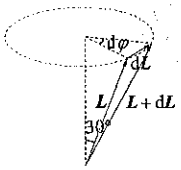
$$a_{Cy}=0$$

$$f=\frac{3}{2}ma_{Cx}$$

$$N=\frac{3}{2}mg$$

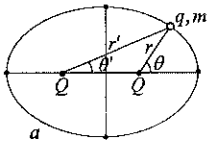
$$\mu_{\min}=\left|\frac{f}{N}\right|=\left|\frac{a_x}{g}\right|=\frac{125\sqrt{3}}{16}\frac{g}{R\omega^2}$$





$$\Omega=\frac{5\sqrt{3}}{2}\frac{g}{R\omega},\quad l_C=\frac{5\sqrt{3}}{6}R$$

题四.



(1) $E_p=\frac{qQ}{4\pi\varepsilon_0}(\frac{1}{r}+\frac{1}{r'})$

$$\left.\begin{array}{l}r+r'=2a\\r-r'=2\Delta\end{array}\right\}\Rightarrow\left\{\begin{array}{l}r=a+\Delta\\r'=a-\Delta\end{array}\right.\Rightarrow E_p=\frac{qQ}{2\pi\varepsilon_0}\frac{a}{a^2-\Delta^2}$$

$\Delta\approx0$, 即 $r'=r$, E_p有极值 $\begin{cases}q,Q\text{同号}, & E_p=E_{p,\min}\\q,Q\text{异号}, & E_p=E_{p,\max}\end{cases}$

$|\Delta|=|\Delta|_{\max}$，即 $r=r_{\min}$ 或 $r_{\max}$，E_p有极值 $\begin{cases}q,Q\text{同号}, & E_p=E_{p,\max}\\q,Q\text{异号}, & E_p=E_{p,\min}\end{cases}$

平衡位置为长轴两端点与短轴两端点；

q,Q同号时， 短轴两端点是稳定平衡点，长轴两端点为不稳定平衡点；

q,Q异号时， 长轴两端点是稳定平衡点，短轴两端点为不稳定平衡点。

(2) 情况 1：q,Q同号， 短轴两端点是稳定平衡点；

$$E_p=\frac{qQ}{4\pi\varepsilon_0}(\frac{1}{r}+\frac{1}{r'})=\frac{qQ}{4\pi\varepsilon_0}(\frac{1}{r}+\frac{1}{2a-r})$$

$E_p=E_p(r)$, $r=r_0=a$, $E_p(r)$有极值

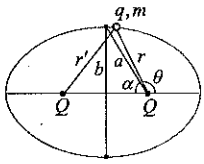
$\Rightarrow$以r为自变量展开

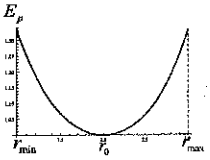
$$r=r_0+\Delta=a+\Delta$$

$$E_p=\frac{qQ}{2\pi\varepsilon_0}\frac{a}{a^2-\Delta^2}\approx\frac{qQ}{2\pi\varepsilon_0 a}(1+\frac{\Delta^2}{a^2})$$

$$=E_{p0}+\frac{qQ(1-\varepsilon^2)^3}{2\pi\varepsilon_0 p^3}\Delta^2$$

$$\left(E_{p0}=\frac{qQ}{2\pi\varepsilon_0 a},\ a=\frac{p}{1-\varepsilon^2}\right)$$





$$E_k=\frac{1}{2}mv^2=\frac{1}{2}m(\dot{r}^2+r^2\dot{\theta}^2)$$

$$\dot{r}=\dot{\Delta}$$

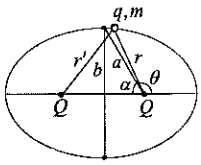
$$\cos\theta=\frac{1}{\varepsilon}(\frac{p}{r}-1)\Rightarrow-\sin\theta\dot{\theta}=-\frac{p}{\varepsilon r^2}\dot{r}=-\frac{p}{\varepsilon r^2}\dot{\Delta}\Rightarrow\dot{\theta}=\frac{p}{\varepsilon r^2\sin\theta}\dot{\Delta}$$

$$E_k=\frac{1}{2}m(1+\frac{p^2}{\varepsilon^2r^2\sin^2\theta})\dot{\Delta}$$

$$\approx\frac{1}{2}m(1+\frac{p^2}{\varepsilon^2a^2\sin^2\alpha})\dot{\Delta}^2$$

$$\approx\frac{1}{2}m(1+\frac{p^2}{\varepsilon^2b^2})\dot{\Delta}^2$$

$$\approx\frac{m}{2\varepsilon^2}\dot{\Delta}^2\left(b=a\sqrt{1-\varepsilon^2},a=\frac{p}{1-\varepsilon^2}\right)$$



$$r=\frac{p}{1+\varepsilon\cos\theta}$$

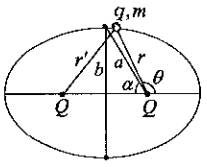
$$E=E_k+E_p=\frac{m}{2\varepsilon^2}\dot{\Delta}^2+\frac{qQ(1-\varepsilon^2)^3}{2\pi\varepsilon_0p^3}\Delta^2+E_{p0}$$

$$\frac{\mathrm{d}E}{\mathrm{d}t}=\frac{m}{\varepsilon^2}\dot{\Delta}\ddot{\Delta}+\frac{qQ(1-\varepsilon^2)^3}{\pi\varepsilon_0p^3}\Delta\dot{\Delta}=0$$

$$\ddot{\Delta}+\frac{qQ(1-\varepsilon^2)^3\varepsilon^2}{\pi\varepsilon_0mp^3}\Delta=0$$

$$\omega=\frac{\varepsilon(1-\varepsilon^2)}{p}\sqrt{\frac{qQ(1-\varepsilon^2)}{\pi\varepsilon_0mp}}$$

$$T=\frac{2\pi}{\omega}=\frac{2\pi p}{\varepsilon(1-\varepsilon^2)}\sqrt{\frac{\pi\varepsilon_0mp}{qQ(1-\varepsilon^2)}}$$



$$E_p=E_{p0}+\frac{qQ(1-\varepsilon^2)^3}{2\pi\varepsilon_0p^3}\Delta^2,E_k=\frac{m}{2\varepsilon^2}\dot{\Delta}^2$$

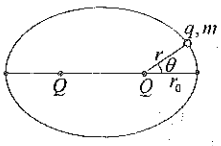
情况 2：q,Q异号，长轴两端点是稳定平衡点：

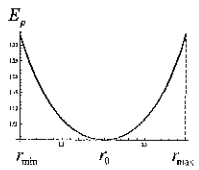
$$E_p=\frac{qQ}{4\pi\varepsilon_0}\left(\frac{1}{r}+\frac{1}{2a-r}\right)$$

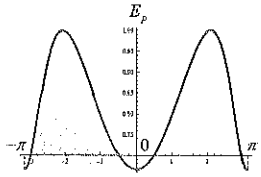
$r=r_0=r_{\min}$不是r的函数$E_p(r)$的极值点

$\theta=\theta_0=0$是θ的函数$E_p(r(\theta))$的极值点

$\Rightarrow$以θ为自变量展开







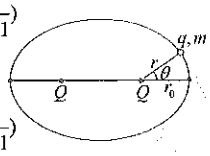
$$r=\frac{p}{1+\varepsilon\cos\theta},\ a=\frac{p}{1-\varepsilon^2}\Rightarrow$$

$$E_p=\frac{qQ}{4\pi\varepsilon_0 p}(1+\varepsilon\cos\theta)\left(1+\frac{1-\varepsilon^2}{\varepsilon^2+2\varepsilon\cos\theta+1}\right)$$

$$\theta=\theta_0+\delta=\delta\ (\delta<<1)$$

$$E_p=\frac{qQ}{4\pi\varepsilon_0 p}(1+\varepsilon\cos\delta)\left(1+\frac{1-\varepsilon^2}{\varepsilon^2+2\varepsilon\cos\delta+1}\right)$$

$$=E_{p0}-\frac{qQ\varepsilon^2}{2\pi\varepsilon_0 p(1+\varepsilon)^2}\delta^2\quad\left(E_{p0}=\frac{qQ}{2\pi\varepsilon_0 p}\right)$$



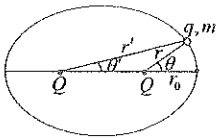
$$E_p=\frac{qQ}{4\pi\varepsilon_0}\left(\frac{1}{r}+\frac{1}{2a-r}\right)$$

$$E_k=\frac{1}{2}mv^2=\frac{1}{2}m(\dot r^2+r^2\dot\theta^2)$$

$$\dot r=\frac{dr}{dt}=\frac{dr}{d\theta}\frac{d\theta}{dt}\approx\left(\frac{dr}{d\theta}\right)_0\dot\delta=0$$

$$r\dot\theta\approx r_0\dot\delta=\frac{p}{1+\varepsilon}\dot\delta$$

$$E_k=\frac{mp^2}{2(1+\varepsilon)^2}\dot\delta^2$$



$$r=\frac{p}{1+\varepsilon\cos\theta}$$

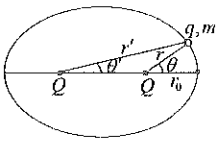
$$E=E_k+E_p=\frac{mp^2}{2(1+\varepsilon)^2}\dot\theta^2-\frac{qQ}{2\pi\varepsilon_0 p}\left(\frac{\varepsilon}{1+\varepsilon}\right)^2\theta^2+E_{p0}$$

$$\frac{dE}{dt}=\frac{mp^2}{(1+\varepsilon)^2}\dot\delta\ddot\delta-\frac{qQ}{\pi\varepsilon_0 p}\left(\frac{\varepsilon}{1+\varepsilon}\right)^2\delta\dot\delta=0$$

$$\ddot\delta-\frac{qQ\varepsilon^2}{\pi\varepsilon_0 mp^3}\delta=0$$

$$\omega=\frac{\varepsilon}{p}\sqrt{\frac{-qQ}{\pi\varepsilon_0 mp}}$$

$$T=\frac{2\pi}{\omega}=\frac{2\pi p}{\varepsilon}\sqrt{\frac{\pi\varepsilon_0 mp}{-qQ}}$$



$$E_k=\frac{mp^2}{2(1+\varepsilon)^2}\dot\delta^2,\ E_p=E_{p0}-\frac{qQ\varepsilon^2}{2\pi\varepsilon_0 p(1+\varepsilon)^2}\delta^2$$

题五.

(1) $U_O=\dfrac{q_{10}}{4\pi\varepsilon_0 r_1}+\dfrac{Q}{4\pi\varepsilon_0 a}=0$

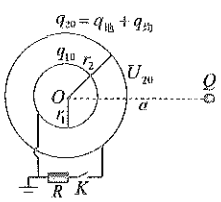
$$q_{10}=-\frac{r_1}{a}Q$$

$$q_{地}=-\frac{r_2}{a}Q$$

$$q_{20}=q_{地}+q_{均}=0\Rightarrow q_{均}=\frac{r_2}{a}Q$$

$$U_{20}=U_{q_{10}}+U_{q_{均}}+\underbrace{U_{q_{地}}+U_Q}_{=0}\qquad(q_{10}\text{均匀分布})$$

$$=\frac{q_{10}}{4\pi\varepsilon_0 r_2}+\frac{q_{均}}{4\pi\varepsilon_0 r_2}=\frac{Q}{4\pi\varepsilon_0 a}\left(1-\frac{r_1}{r_2}\right)$$



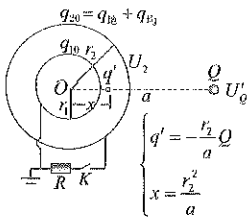
$$U'_Q=U_{q_{10}}+U_{q_{均}}+U_{q_{地}}$$

$$=U_{q_{10}}+U_{q_{均}}+U_{q'}$$

$$=\frac{q_{10}}{4\pi\varepsilon_0 a}+\frac{q_{均}}{4\pi\varepsilon_0 a}+\frac{q'}{4\pi\varepsilon_0(a-x)}$$

$$=-\frac{Q}{4\pi\varepsilon_0 a^2}\left(r_1+\frac{r_2^3}{a^2-r_2^2}\right)$$

$$\begin{cases}q'=-\dfrac{r_2}{a}Q\\ x=\dfrac{r_2^2}{a}\end{cases}$$

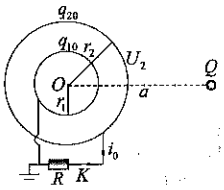


$$W_e=\frac{1}{2}QU'_Q=-\frac{Q^2}{8\pi\varepsilon_0 a^2}\left(r_1+\frac{r_2^3}{a^2-r_2^2}\right)$$

$$q_{均}=\frac{r_2}{a}Q,\ q_{10}=-\frac{r_1}{a}Q$$

(2)

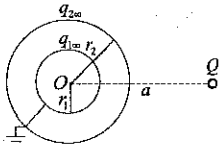
$$i_0=\frac{U_2}{R}=\frac{Q}{4\pi\varepsilon_0 aR}(1-\frac{r_1}{r_2})$$



(3)

$$q_{1\infty}=0$$

$$q_{2\infty}=q_{地}=-\frac{r_2}{a}Q$$



$$U_2=\frac{Q}{4\pi\varepsilon_0 a}(1-\frac{r_1}{r_2}),\quad q_{地}=-\frac{r_2}{a}Q$$

(4) $q_{地}=-\frac{r_2}{a}Q$

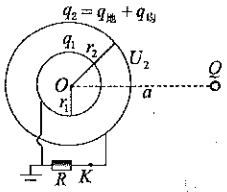
$$q_2=q_{地}+q_{均}\Rightarrow q_{均}=q_2+\frac{r_2}{a}Q$$

$$U_O=\frac{q_1}{4\pi\varepsilon_0 r_1}+\frac{q_2}{4\pi\varepsilon_0 r_2}+\frac{Q}{4\pi\varepsilon_0 a}=0$$

$$q_1=-(\frac{r_1}{r_2}q_2+\frac{r_1}{a}Q)$$

$$U_2=U_{q_1}+U_{q_{均}}+\underbrace{U_{q_{地}}+U_Q}_{=0}\quad (q_1均匀分布)$$

$$=\frac{q_1}{4\pi\varepsilon_0 r_2}+\frac{q_{均}}{4\pi\varepsilon_0 r_2}=\frac{1}{4\pi\varepsilon_0}(1-\frac{r_1}{r_2})(\frac{q_2}{r_2}+\frac{Q}{a})$$



$$i=\frac{U_2}{R}=\frac{1}{4\pi\varepsilon_0 R}(1-\frac{r_1}{r_2})(\frac{q_2}{r_2}+\frac{Q}{a})$$

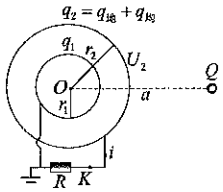
$$\frac{di}{dt}=\frac{1}{4\pi\varepsilon_0 r_2 R}(1-\frac{r_1}{r_2})\frac{dq_2}{dt}$$

$$=-\frac{1}{4\pi\varepsilon_0 r_2 R}(1-\frac{r_1}{r_2})i\quad (i=-\frac{dq_2}{dt})$$

$$\frac{di}{i}=-\frac{1}{4\pi\varepsilon_0 r_2 R}(1-\frac{r_1}{r_2})dt\equiv-\gamma dt$$

$$\int_{i_0}^{i}\frac{di}{i}=-\int_0^t\gamma dt$$

$$i=i_0e^{-\gamma t}\quad\left[i_0=\frac{Q}{4\pi\varepsilon_0 aR}(1-\frac{r_1}{r_2}),\gamma=\frac{1}{4\pi\varepsilon_0 r_2 R}(1-\frac{r_1}{r_2})\right]$$

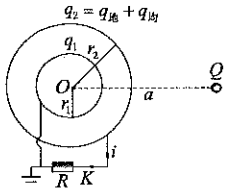


$$i_0=\frac{Q}{4\pi\varepsilon_0 aR}(1-\frac{r_1}{r_2}),U_2=\frac{1}{4\pi\varepsilon_0}(1-\frac{r_1}{r_2})(\frac{q_2}{r_2}+\frac{Q}{a})$$

(5)

$$Q_J=\int_0^\infty i^2Rdt=\int_0^\infty i_0^2Re^{-2\gamma t}dt$$

$$=\frac{i_0^2R}{2\gamma}=\frac{Q^2}{8\pi\varepsilon_0 a^2}(r_2-r_1)$$



$$i=i_0e^{-\gamma t}\left[i_0=\frac{Q}{4\pi\varepsilon_0 aR}(1-\frac{r_1}{r_2}),\gamma=\frac{1}{4\pi\varepsilon_0 r_2 R}(1-\frac{r_1}{r_2})\right]$$

另解：

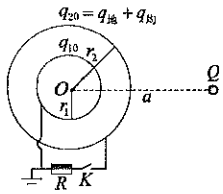
$$W_e=-\frac{Q^2}{8\pi\varepsilon_0 a^2}(r_1+\frac{r_2^3}{a^2-r_2^2})$$

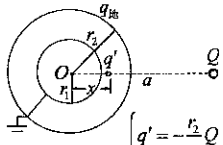
$$U'_Q=U_{q_{地}}=U_{q'}=\frac{q'}{4\pi\varepsilon_0(a-x)}$$

$$=-\frac{r_2Q}{4\pi\varepsilon_0(a^2-r_2^2)}$$

$$W_{e\infty}=\frac{1}{2}QU'_Q=-\frac{r_2Q^2}{8\pi\varepsilon_0(a^2-r_2^2)}$$

$$Q_J=W_{e0}-W_{e\infty}=\frac{Q^2}{8\pi\varepsilon_0 a^2}(r_2-r_1)$$





$$t\to\infty\quad\begin{cases}q'=-\frac{r_2}{a}Q\\ x=\frac{r_2^2}{a}\end{cases}$$

$$W_e=-\frac{Q^2}{8\pi\varepsilon_0 a^2}(r_1+\frac{r_2^3}{a^2-r_2^2})$$

题六.

(1)卡诺定理的证明：

二热源热机循环：

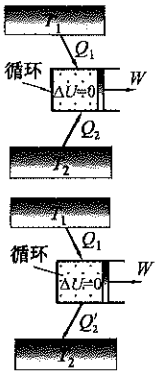
$$\sum_{i=1}^{n}\frac{Q_i}{T_i}\le 0\Rightarrow\frac{Q_1}{T_1}+\frac{Q_2}{T_2}\le 0$$

Q'_2表示向低温热源T_2放出的热量：$Q_2=-Q'_2$

$$\frac{Q_1}{T_1}-\frac{Q'_2}{T_2}\le 0\Rightarrow\frac{Q_1}{T_1}\le\frac{Q'_2}{T_2}\Rightarrow\frac{Q'_2}{Q_1}\ge\frac{T_2}{T_1}$$

热机的效率：

$$\eta=\frac{W}{Q_1}=1-\frac{Q'_2}{Q_1}\le 1-\frac{T_2}{T_1}$$

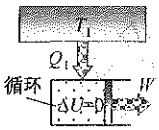


热力学第二定律开尔文描述的证明：

反证法：假设开尔文描述不成立，则

$$Q_1 > 0 \Rightarrow \frac{Q_1}{T_1} > 0$$

违背克劳修斯不等式：$\sum_{i=1}^{n}\frac{Q_i}{T_i} \le 0$



热力学第二定律克劳修斯描述的证明：

反证法：假设克劳修斯描述不成立，则

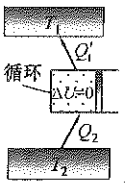
$T_1 > T_2$时，$Q_1' = Q_2 > 0$

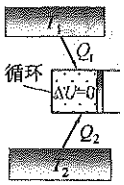
$$\frac{Q_1'}{T_1} < \frac{Q_2}{T_2}$$

Q_1表示从高温热源T_2吸收的热量：$Q_1' = -Q_1$

$$-\frac{Q_1}{T_1} < \frac{Q_2}{T_2} \Rightarrow \frac{Q_1}{T_1} + \frac{Q_2}{T_2} > 0$$

违背克劳修斯不等式：$\sum_{i=1}^{n}\frac{Q_i}{T_i} \le 0$





(2)

$$\sum_{i=1}^{n}\frac{Q_i}{T_i} \le 0$$

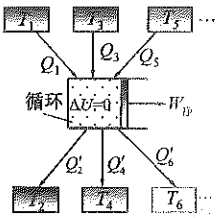
$$\sum_{j=1}\frac{Q_{2j-1}}{T_{2j-1}} - \sum_{k=1}\frac{Q'_{2k}}{T_{2k}} \le 0,\ \sum_{j=1}\frac{Q_{2j-1}}{T_{2j-1}} \le \sum_{k=1}\frac{Q'_{2k}}{T_{2k}}$$

$$T_{2j-1} \le T_1 \Rightarrow \sum_{j=1}\frac{Q_{2j-1}}{T_{2j-1}} \ge \frac{1}{T_1}\sum_{j=1}Q_{2j-1} = \frac{Q_{吸}}{T_1}$$

$$T_k \ge T_2 \Rightarrow \sum_{k=1}\frac{Q'_{2k}}{T_{2k}} \le \frac{1}{T_2}\sum_{k=1}Q'_{2k} = \frac{Q_{放}}{T_2}$$

$$\frac{Q_{吸}}{T_1} \le \sum_{j=1}\frac{Q_{2j-1}}{T_{2j-1}} \le \sum_{k=1}\frac{Q'_{2k}}{T_{2k}} \le \frac{Q_{放}}{T_2},\ \frac{Q_{吸}}{T_1} \le \frac{Q_{放}}{T_2},\ \frac{Q_{放}}{Q_{吸}} \ge \frac{T_2}{T_1}$$

$$\eta = \frac{W_{净}}{Q_{吸}} = 1 - \frac{Q_{放}}{Q_{吸}} \le 1 - \frac{T_2}{T_1}$$

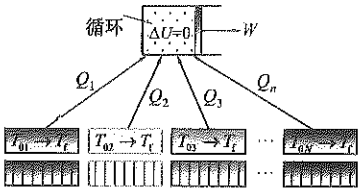


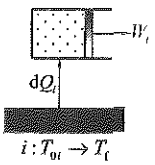
(3)

$$\left.\begin{aligned} W &= Q_1 + Q_2 + \cdots + Q_N = \sum_{i=1}^{N} Q_i \\ Q_i &= C(T_{0i} - T_f) \end{aligned}\right\} \Rightarrow W = NC\left(\frac{\sum_{i=1}^{N} T_{0i}}{N} - T_f\right)$$

$$\left.\begin{aligned} &\sum_{j=1}^{n}\frac{Q_j}{T_j} \le 0 \to \sum_{i=1}^{N}\int\frac{dQ_i}{T_i} \le 0 \\ &\int\frac{dQ_i}{T_i} = \int_{T_{0i}}^{T_f}\frac{-CdT_i}{T_i} = -C\ln\frac{T_f}{T_{0i}} \end{aligned}\right\} \Rightarrow \sum_{i=1}^{N}\ln\frac{T_f}{T_{0i}} \ge 0 \Rightarrow \frac{T_f^N}{\prod_{i=1}^{N}T_{0i}} \ge 1$$

$$\Rightarrow T_f \ge \left(\prod_{i=1}^{N}T_{0i}\right)^{1/N}$$





$$W \le NC\left[\frac{\sum_{i=1}^{N}T_{0i}}{N} - \left(\prod_{i=1}^{N}T_{0i}\right)^{1/N}\right]$$

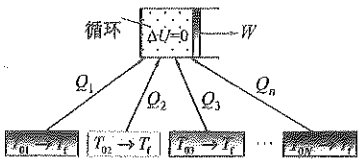
$$W_{max} = NC\left[\frac{\sum_{i=1}^{N}T_{0i}}{N} - \left(\prod_{i=1}^{N}T_{0i}\right)^{1/N}\right] = NC(\bar{T}_0 - \tilde{T}_0)$$

$\bar{T}_0 = \frac{\sum_{i=1}^{N}T_{0i}}{N}$ ——算术平均值

$\tilde{T}_0 = \left(\prod_{i=1}^{N}T_{0i}\right)^{1/N}$ ——几何平均值

$$W = NC\left(\frac{\sum_{i=1}^{N}T_{0i}}{N} - T_f\right)$$

$$T_f \ge \left(\prod_{i=1}^{N}T_{0i}\right)^{1/N}$$



题七.

(1) $\sin i = n\sin r$ (1)

$n\sin i' = \sin r'$

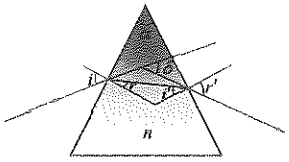
$$\begin{cases} \alpha = r + i' \Rightarrow i' = -r + \alpha \\ \delta = (i - r) + (r' - i') = i + r' - \alpha \\ \qquad \Rightarrow r' = -i + \delta + \alpha \end{cases}$$

$n\sin(-r+\alpha) = \sin(-i+\delta+\alpha)$

$$\begin{cases} \alpha << 1 \Rightarrow \delta << 1 \\ \sin(-r+\alpha) \approx -\sin r + \cos r\cdot\alpha \\ \sin(-i+\delta+\alpha) \approx -\sin i + \cos i\cdot(\delta+\alpha) \end{cases}$$

$-n\sin r + n\cos r\cdot\alpha \approx -\sin i + \cos i\cdot(\delta+\alpha)$ (2)

$$(1)(2) \Rightarrow \delta = \left(\frac{n\cos r}{\cos i} - 1\right)\alpha = \left(\frac{\sqrt{n^2 - \sin^2 i}}{\cos i} - 1\right)\alpha$$



(2)

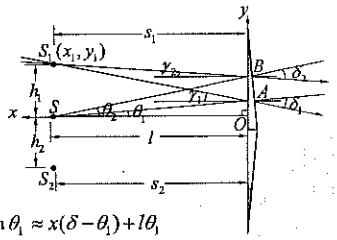
$i_1=\theta_1<<1\Rightarrow\delta_1=(n-1)\alpha$

$i_2=\theta_2<<1\Rightarrow\delta_2\approx(n-1)\alpha$

$\delta_1=\delta_2=(n-1)\alpha\equiv\delta$

$\gamma_1=\delta_1-\theta_1=\delta-\theta_1$

$\gamma_2=\delta_2-\theta_2=\delta-\theta_2$



$AS_1: y=x\tan(\delta_1-\theta_1)+l\tan\theta_1\approx x(\delta-\theta_1)+l\theta_1$

$BS_1: y=x\tan(\delta_2-\theta_2)+l\tan\theta_2\approx x(\delta-\theta_2)+l\theta_2$

$$\begin{cases}y_1=x_1(\delta-\theta_1)+l\theta_1\\ y_1=x_1(\delta-\theta_2)+l\theta_2\end{cases}\Rightarrow\begin{cases}x_1=l\\ y_1=l\delta=(n-1)\alpha l\end{cases}$$ 与θ_1、θ_2无关！

$s_1=l, h_1=(n-1)\alpha l$

同理：

$s_2=l, h_2=(n-1)\alpha l$

$\delta=(\frac{\sqrt{n^2-\sin^2 i}}{\cos i}-1)\alpha$

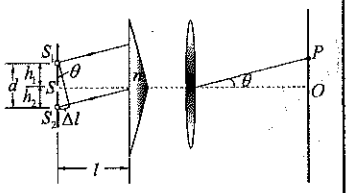
(3)

$d=h_1+h_2=2(n-1)\alpha l$

$\Delta l=l_2-l_1=d\sin\theta$

$=2(n-1)\alpha l\sin\theta=k\lambda$

$\sin\theta=k\frac{\lambda}{2(n-1)\alpha l}$



$\Delta l=l_2-l_1=2(n-1)\alpha l\sin$

$h_1=h_2=(n-1)\alpha l$

(4)

设想双棱镜切去厚度为e的薄片，则屏幕上出现明条纹的条件与第（3）问的相同。

当存在厚度为e的薄片时，从虚光源S_2发出的光产生了$(n-1)e$的附加光程。

$e=\frac{b}{\cos\alpha}\sin 2\alpha\approx 2\alpha b$

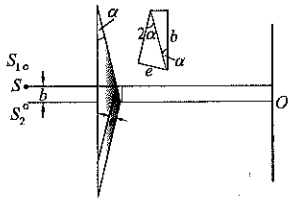
$l_2'=l_2+(n-1)e=l_2+2(n-1)\alpha b$

$\Delta l'=l_2'-l_1=l_2-l_1+2(n-1)\alpha b$

$=2(n-1)\alpha(l\sin\theta+b)=k\lambda$

$\sin\theta=\frac{k\lambda}{2(n-1)l\alpha}-\frac{b}{l}$

$(k=0,1,2,\cdots)$



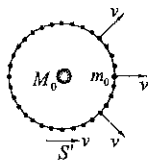
$\Delta l=l_2-l_1=2(n-1)\alpha l\sin\theta$

题八.

(1) 质心参照系(S')：

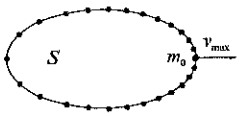
$M_0=\frac{Nm_0}{\sqrt{1-\beta^2}}\ (\beta=\frac{v}{c})$

$m_0=\frac{M_0}{N}\sqrt{1-\beta^2}$



实验室参照系(S)：

$v_{max}=\frac{v+v}{1+v^2/c^2}=\frac{2v}{1+\beta^2}$



$E_{k,max}=m_{max}c^2-m_0c^2=\frac{m_0c^2}{\sqrt{1-v_{max}^2/c^2}}-m_0c^2$

$=\frac{2\beta^2}{1-\beta^2}m_0c^2=\frac{2\beta^2}{\sqrt{1-\beta^2}}\frac{M_0}{N}c^2$

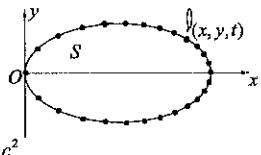
(2)

假设$t=t'=0$时刻S和S'坐标原点重合，此时引爆飞弹。

S：t时刻任一弹片的位置为(x,y)

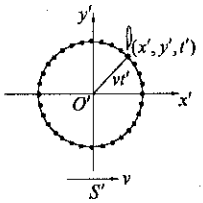
S：t时刻在(x,y)处发生一事件，

其时空坐标为：$P(x,y,t)$



S'：这一事件的位置时空坐标：

$x'=\frac{x-vt}{\sqrt{1-\beta^2}},\ y'=y,\ t'=\frac{t-vx/c^2}{\sqrt{1-\beta^2}}$



S'：(x',y',t')满足：

$x'^2+y'^2=(vt')^2$

$(\frac{x-vt}{\sqrt{1-\beta^2}})^2+y^2=(\frac{t-vx/c^2}{\sqrt{1-\beta^2}})^2$

$(1+\beta^2)x^2-2vtx+y^2=0$

(3)

分析：

$dn=\lambda dl$

$dn'=\frac{N}{2\pi}d\theta'$

$dn=dn'\Rightarrow\lambda=\frac{N}{2\pi}\frac{d\theta'}{dl}$

$dl\sim d\theta'$?

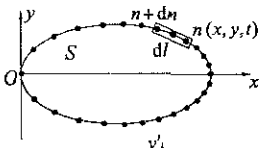
$dl=\sqrt{(dx)^2+(dy)^2}$

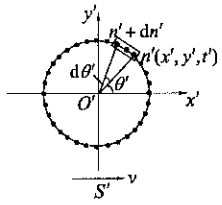
$dx,dy\sim d\theta'$?

$x,y\sim\theta'$?

$x'=vt'\cos\theta', y'=vt'\sin\theta'$

$x,y,t\sim x',y',t'$?





(3) 考察 S' 中沿 θ' 方向射出的弹片 P：

S' : t' 时刻弹片 P 的坐标：

$$x' = vt'\cos\theta',\ y' = vt'\sin\theta'$$

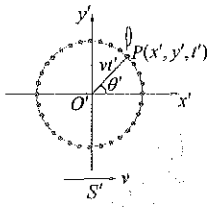
S : t' 时刻该弹片所在处发生一事件，其时空坐标为：$P(x', y', t')$

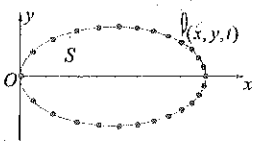
S：这一事件的时空坐标：

$$\begin{cases} x = \dfrac{x' + vt'}{\sqrt{1-\beta^2}} = \dfrac{1+\cos\theta'}{\sqrt{1-\beta^2}}vt' \\ y = y' = vt'\sin\theta' \\ t = \dfrac{t' + vx'/c^2}{\sqrt{1-\beta^2}} = \dfrac{1+\beta^2\cos\theta'}{\sqrt{1-\beta^2}}t' \end{cases}$$

S : t 时刻的弹片 P 的坐标：

$$x = \frac{1+\cos\theta'}{1+\beta^2\cos\theta'}vt,\quad y = \frac{\sin\theta'\sqrt{1-\beta^2}}{1+\beta^2\cos\theta'}vt$$





考察 S' 中沿 $\theta' \sim \theta' + \mathrm{d}\theta'$ 方向射出的 $\mathrm{d}n'$ 个弹片：

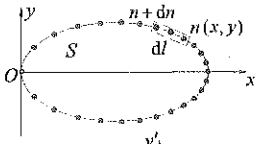
S : t 时刻这些弹片头尾的距离：

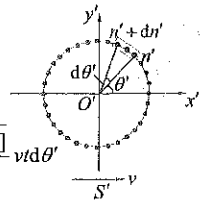
$$\mathrm{d}x = -\frac{(1-\beta^2)\sin\theta'}{(1+\beta^2\cos\theta')^2}vt\mathrm{d}\theta'$$

$$\mathrm{d}y = \frac{\sqrt{1-\beta^2}(\beta^2+\cos\theta')}{(1+\beta^2\cos\theta')^2}vt\mathrm{d}\theta'$$

$$\mathrm{d}l = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}y)^2}$$

$$= \frac{\sqrt{(1-\beta^2)[(\beta^2+\cos\theta')^2 + (1-\beta^2)\sin^2\theta']}}{(1+\beta^2\cos\theta')^2}vt\mathrm{d}\theta'$$



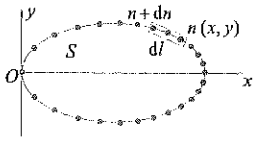


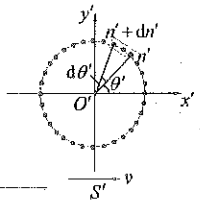
$$x = \frac{1+\cos\theta'}{1+\beta^2\cos\theta'}vt,\ y = \frac{\sin\theta'\sqrt{1-\beta^2}}{1+\beta^2\cos\theta'}vt,\ t' = \frac{\sqrt{1-\beta^2}}{1+\beta^2\cos\theta'}t,$$

$$\mathrm{d}n = \mathrm{d}n' = \frac{N}{2\pi}\mathrm{d}\theta'$$

$$\lambda = \frac{\mathrm{d}n}{\mathrm{d}l} = \frac{N}{2\pi vt}\frac{(1+\beta^2\cos\theta')^2}{\sqrt{(1-\beta^2)[(\beta^2+\cos\theta')^2 + (1-\beta^2)\sin^2\theta']}}$$

$$\lambda_{\max} = \lambda\big|_{\theta'=0} = \frac{N}{2\pi vt}\frac{1+\beta^2}{\sqrt{1-\beta^2}}$$





$$\mathrm{d}l = \frac{\sqrt{(1-\beta^2)[(\beta^2+\cos\theta')^2 + (1-\beta^2)\sin^2\theta']}}{(1+\beta^2\cos\theta')^2}vt\mathrm{d}\theta'$$